MODULE NAME:	MODULE CODE:
PROGRAMMING FOR DATA ANALYTICS 1	PDAN8411/w

Formative 2 Part 2

Student Details	
Name: Sifiso	**Surname:** Mahlangu
Student Number: ST10545223	**Module Code:** PDAN8411/w
Date: 21 October 2026	

Task Description
Part 2: Classification and Model Improvement

1. Choice of dataset

I used the **Breast Cancer Wisconsin (Diagnostic)** dataset from the UCI Machine Learning Repository (Wolberg et al., 1993). Each of its 569 rows is one patient's breast mass. Thirty numeric features describe the cell nuclei in a digitized image of a fine needle aspirate (FNA), a routine and minimally invasive biopsy, and each mass carries a confirmed diagnosis of malignant or benign (Street, Wolberg and Mangasarian, 1993).

Criterion	Why this dataset meets it
Provenance	Collected at the University of Wisconsin Hospitals, documented in a peer-reviewed paper and published under a DOI on the UCI repository with a CC BY 4.0 licence (Wolberg et al., 1993).
Completeness	No missing values, no duplicate rows and no impossible (negative) values. Confirmed in the notebook, not just taken from the documentation.
Label quality	Every diagnosis is confirmed, so the target is ground truth rather than an estimate.
Fit to the brief	The brief asks for a cancer classifier. A binary malignant/benign target maps straight onto the scheme's decision: fast-track the claim or not.
Fit to the method	All 30 features are continuous measurements, which suits logistic regression and p-value feature selection without heavy encoding.
Size and balance	569 cases at a 63/37 benign/malignant split: enough for a train/test split with cross-validation, and not so imbalanced that resampling is needed.

What a classification dataset needs, and how this one measures up. Supervised classification needs a labelled target with a fixed set of classes, here malignant or benign. It needs input features that are measured before the decision is made, so the model can be used on new cases; these are biopsy measurements, available before a claim is assessed. It needs enough examples of each class to learn from and to test on, and it needs features in a form the algorithm can use. Logistic regression needs numeric inputs, and all 30 features are continuous, so no encoding is needed beyond the target (James et al., 2021).

Common dataset pitfalls I checked for:

Pitfall	Present here?	How I handled it
Raw file not analysis-ready	Yes: no header row, an ID column, text labels (M/B)	Named the columns from the documentation, dropped the ID, encoded M = 1, then verified against an independent copy
Missing, duplicate or impossible values	No	Checked rather than assumed (notebook section 3.2)
Class imbalance	Moderate: 63/37	Stratified split; recall, F2 and PR-AUC instead of accuracy; tested balanced class weights
Multicollinearity	Severe: 44 feature pairs above r = 0.8	Correlation filter and VIF before p-value selection
Outliers	Yes, mostly malignant	Kept as genuine cases; scaled and regularized
Data leakage	Risk if features chosen on all data	Split first; all selection and tuning on training data only

Pitfall	Present here?	How I handled it
Small sample	569 rows; 42 malignant in test	Cross-validation for every decision; test set used once
Representativeness	One US hospital, early 1990s	Flagged as a limitation; revalidate on scheme data
Privacy and licence	Anonymized IDs; CC BY 4.0	Fine for study; real member data falls under POPIA (South Africa, 2013)

Limitation: the data comes from one US hospital and covers breast masses only. It is a proof of concept for the scheme's process; the model would need retraining and validation on the scheme's own member data before real use.

2. Choice of algorithm

I chose **logistic regression**. It models the probability that a mass is malignant as p = 1 / (1 + e^-(b0 + b1x1 + ... + bkxk)), and a case is flagged when p passes a chosen threshold (James et al., 2021).

Requirement from the brief	How logistic regression meets it
Decisions about members' claims must be explainable	Each coefficient converts to an odds ratio, so the scheme can say which measurements drove a flag. Tree ensembles and neural networks can't do this as directly.
Feature selection with p-values is suggested	Logistic regression produces coefficient p-values (Hosmer, Lemeshow and Sturdivant, 2013); most other classifiers don't.
Speed up benefit approval	It outputs a probability, so the threshold can be moved to catch more cancers. Training and scoring take milliseconds.
Small dataset	Few parameters, so less risk of overfitting 455 training rows than a high-capacity model.

Why the alternatives are less suitable. I didn't just argue this; I ran five common classifiers through the same 5-fold cross-validation, on the same six features and training data (notebook section 8.1).

Algorithm	CV recall	CV F2	CV ROC-AUC	Why it's less suitable for this brief
Logistic regression (chosen)	0.9706	0.9683	0.9960	Best recall and F2, and fully explainable
Random forest	0.9412	0.9444	0.9924	Lower recall; explains itself only through feature importances, not per-feature odds
Support vector machine (RBF)	0.9353	0.9407	0.9959	Ranks well but no probabilities by default and no interpretable coefficients
k-nearest neighbours	0.9059	0.9186	0.9850	Misses more cancers; must store all training data and gives no explanation
Gaussian naive Bayes	0.9059	0.9163	0.9906	Assumes features are independent, which the correlated nuclei measurements break
Decision tree	0.9059	0.9060	0.9266	Overfits a dataset this size; weakest ranking

To be fair to the others, logistic regression ran with its tuned settings while they ran on defaults, so this isn't proof that it's the best possible model. It does show that nothing off the shelf clearly beats it, so there is no accuracy reason to give up its interpretability. Gradient boosting and neural networks were not tested: they tend to pay off on larger datasets than 569 rows and are the hardest to explain to a member.

3. Analysis plan

Step	Plan	Output
a. EDA	Check shape, types, missing values, duplicates, negative values and zeros. Class balance. Histograms and boxplots by diagnosis. IQR outlier count and skewness. Correlation heatmap and correlation with the target.	Clean, understood data; decisions on scaling, outliers and stratification
b. Feature selection	Split 80/20 (stratified) first. On the training set only: correlation filter at |r| > 0.8, VIF check (target below 10), then backward elimination dropping the highest p-value until all are below 0.05.	A small, non-redundant, significant feature set
c. Train	Pipeline of StandardScaler and LogisticRegression. Baseline uses defaults: C = 1.0, L2 penalty, lbfgs solver, no class weights, threshold 0.5, max_iter = 1000.	Baseline model
d. Evaluate	Recall first, then precision, specificity, F1, F2, ROC-AUC and the confusion matrix. 5-fold stratified CV on training data plus one test-set score. Compare against an always-benign model.	Clear view of where the model falls short
Retrain	Grid search over C and class_weight scored on F2; pick the threshold from out-of-fold predictions. Test set used once, at the end.	Improved final model
e. Report	Data and cleaning, EDA visuals, feature selection, training, metric choice, results, retraining, limitations.	This report and the notebook

Risk I planned for	Safeguard
Test set leaking into choices	Split first; feature selection, tuning and threshold all on training data via cross-validation
Accuracy hiding missed cancers	Recall-led metrics; compare against an always-benign model
Unstable p-values from correlated features	Correlation filter and VIF before backward elimination
Model working only on paper	Benchmark other algorithms; worked test case; hand-check metrics against scikit-learn
Misuse on real members	Fast-track only, never auto-decline; POPIA for any real data

4. Report to the medical scheme

4.1 What the data contains

Ten properties of the cell nuclei were measured on each image: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry and fractal dimension. Each is summarized three ways, the mean, the standard error ("error") and the mean of the three largest values ("worst"), giving 30 features (Street, Wolberg and Mangasarian, 1993). The target is the diagnosis.

I downloaded the raw **wdbc.data** file from the UCI dataset page. It has no header row, a patient ID in the first column and the diagnosis as text (M or B), so it needed preparing before any analysis. As an integrity check I compared every value against the independent copy that ships with scikit-learn: all 569 x 30 values, the column order and the labels matched exactly.

4.2 Data cleaning

Check	Result	Action
Header row	None in the raw file	Named 32 columns in the documented order
Patient ID	569 unique IDs, no medical meaning	Dropped so the model can't learn from record numbers
Diagnosis labels	Only M and B found	Encoded M = 1 (positive class), B = 0
Integrity	Matches scikit-learn's copy exactly	Confirms download, column order and encoding
Missing values	0	None needed
Duplicate rows	0	None needed
Negative values	0	None needed
Zeros	13 rows, only in the six concavity and concave-points columns; all 13 benign	Kept: a perfectly smooth nucleus has zero concavity
Outliers (1.5 x IQR rule)	171 rows have at least one; all 65 high outliers in area error are malignant	Kept: they are real, severe cases, not errors
Scale	Features range from fractions of 1 (smoothness) to over 2,000 (worst area)	Standardized inside the model pipeline

The decision to keep the outliers matters. Deleting them would remove the most severe malignant cases, exactly the ones the scheme most needs to catch.

4.3 What the exploratory analysis showed

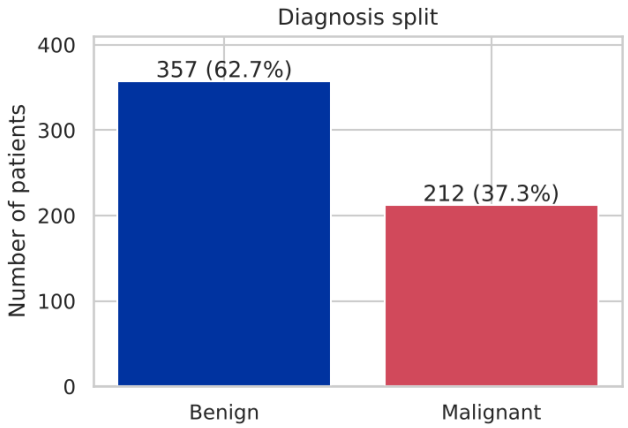


Figure 1: 357 benign (62.7%) and 212 malignant (37.3%) cases

The split is moderately imbalanced. A model that always says benign would already be about 63% accurate while catching no cancers, so **accuracy can't be the main measure of success**. I stratified the train/test split so both sets keep this ratio.

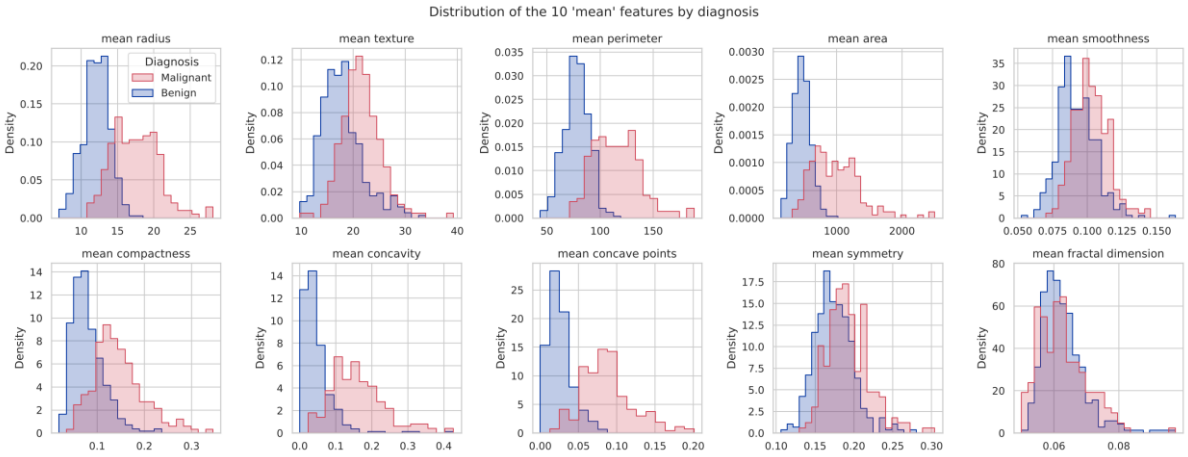


Figure 2: Distribution of the ten 'mean' features by diagnosis

Malignant masses are clearly larger (radius, perimeter, area) and more irregular (concavity, concave points). Smoothness, symmetry and fractal dimension overlap heavily between the groups, so on their own they add little. Figure 3 shows the separation for four of the strongest features.

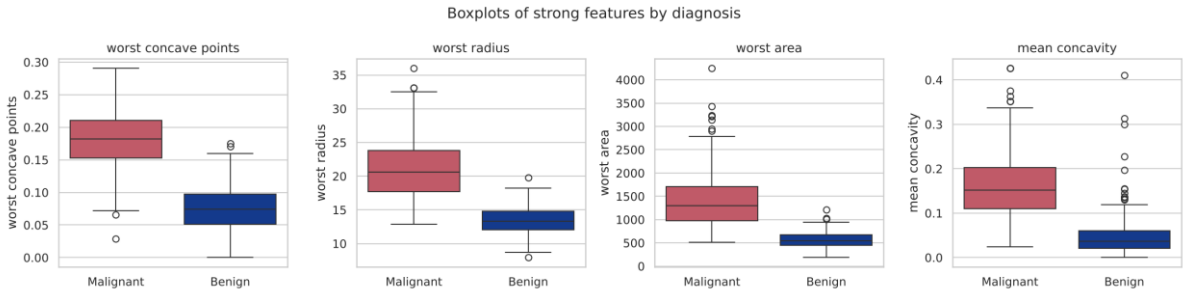


Figure 3: Boxplots of strong features by diagnosis

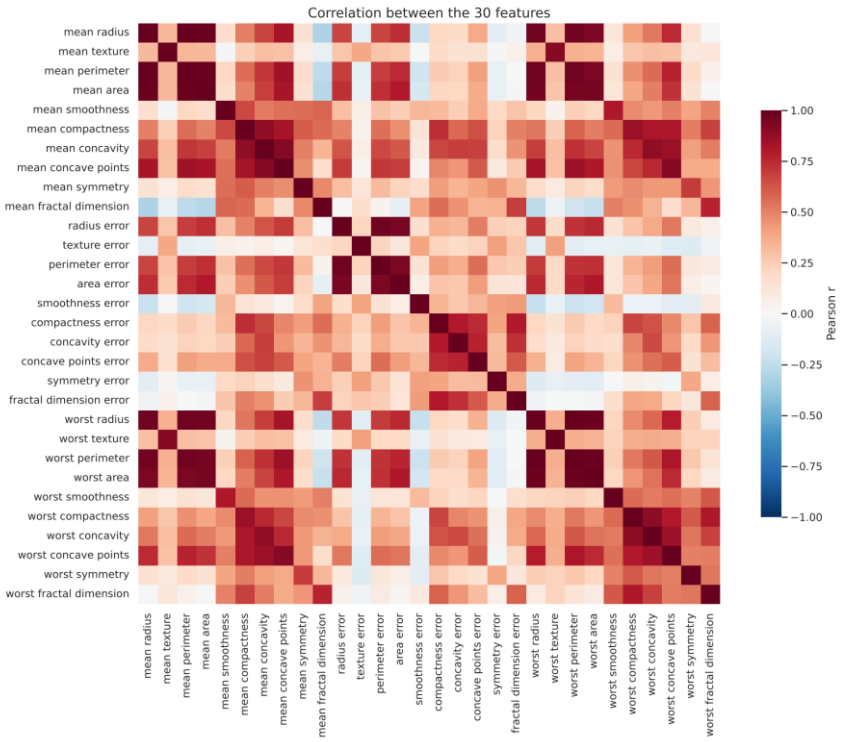


Figure 4: Correlation between all 30 features

The heatmap shows heavy **multicollinearity**, meaning several features carry the same information. Of the 435 feature pairs, 44 correlates above 0.8 and 15 above 0.95. Radius, perimeter and area are near copies of one another, as are the mean and worst versions of a property. This makes logistic regression coefficients and p-values unstable (James et al., 2021), so it shaped the feature selection.

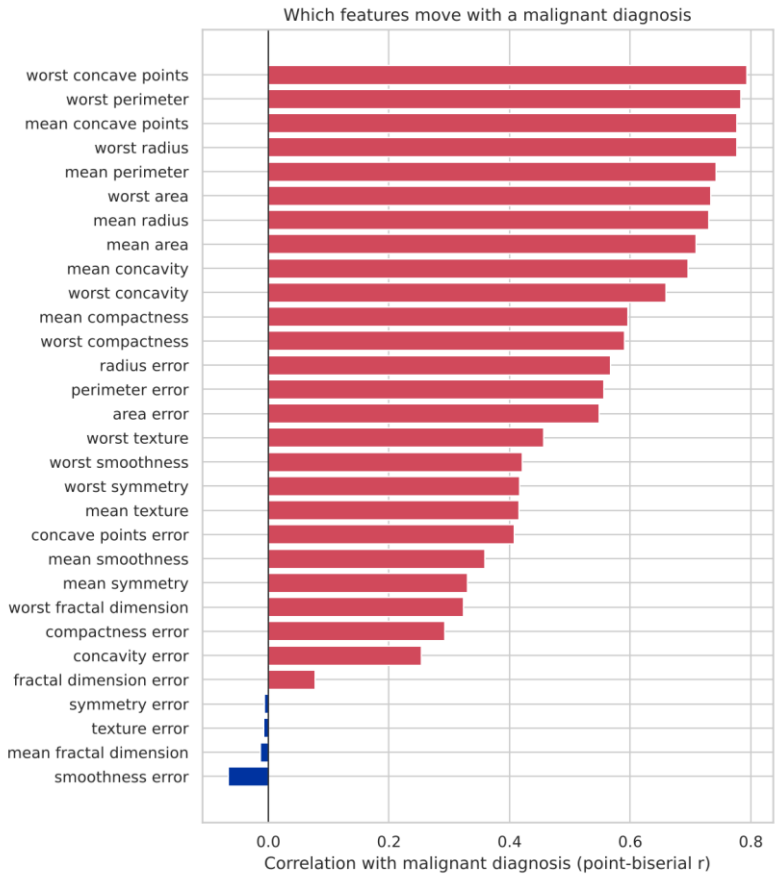


Figure 5: Correlation of each feature with a malignant diagnosis

The features most associated with malignancy are worst concave points (r = 0.794), worst perimeter (0.783), mean concave points (0.777) and worst radius (0.776).

How the EDA shaped the model. Each finding fed directly into a training decision:

EDA finding	Decision it drove
63/37 class split	Stratified split, recall-focused metrics, balanced class weights tested
Features on very different scales	StandardScaler inside the pipeline
Outliers are mostly severe malignant cases	Kept; scaling and regularization limit their pull
Heavy multicollinearity	Correlation filter and VIF before p-value selection
Size and concavity separate the classes best	Expected to survive selection, and they did (section 4.4)

4.4 Feature selection

I split the data first (455 training and 114 test patients, 42 of them malignant) and selected features using the training set only. Selecting on all the data would let the test set influence the model, a form of data leakage that inflates test scores (Muller and Guido, 2016).

Step	Method	Result
1. Correlation filter	Rank by correlation with the diagnosis; keep a feature only if $\|r\| < 0.8$ with every feature already kept	30 down to 14 features

Step	Method	Result
2. VIF check	VIF = 1 / (1 - R squared); above 10 signals a problem	Max VIF fell from 4,110.57 (mean perimeter) to 14.17
3. Backward elimination	statsmodels Logit (Seabold and Perktold, 2010); drop the highest p-value above 0.05 and refit	8 features dropped; 6 remain

Final feature	Coefficient (statsmodels, standardized)	p-value
worst concave points	5.0235	< 0.001
worst radius	3.8526	0.0020
radius error	5.8354	0.0011
worst texture	3.6145	< 0.001
compactness error	-1.0948	0.0303
texture error	-1.5817	0.0458

All six are significant at the 5% level, the highest VIF is now 5.65, and McFadden's pseudo R-squared is 0.907. As a check, logistic regression on all 30 features scored exactly the same on the test set (accuracy 0.9649, recall 0.9286), so dropping 24 features cost nothing and made the model far easier to explain.

4.5 How the model was trained

Scaling and the model sit in one scikit-learn pipeline (Pedregosa et al., 2011), so the scaler learns its means and standard deviations from training data only, including within each cross-validation fold. The baseline used the defaults: C = 1.0, L2 penalty, lbfgs solver, no class weighting and a 0.5 threshold. The only change was max_iter = 1000, so the solver always converges.

4.6 How the results were interpreted

The two possible errors cost very different amounts. A **false negative** means a member with cancer isn't fast-tracked, the exact problem the scheme wants to fix. A **false positive** means a clinical reviewer checks a claim that turns out benign, which costs some reviewer time. Muller and Guido (2016) make the same point for cancer screening: the metric must reflect which mistake matters more.

Metric	Formula	Role here
Recall (sensitivity)	TP / (TP + FN)	Primary: share of real cancers caught
Precision	TP / (TP + FP)	Reviewer workload: share of flags that are real
Specificity	TN / (TN + FP)	Share of benign cases correctly cleared
F1 / F2	Harmonic mean of precision and recall; F2 weights recall twice as heavily	F2 is the single score used for tuning
ROC-AUC	Area under the ROC curve	How well the model ranks malignant above benign

Metric	Formula	Role here
PR-AUC (average precision)	Area under the precision-recall curve	Ranking quality focused on the malignant class
Accuracy	(TP + TN) / total	Reported, but misleading on its own here

ROC-AUC alone can look flattering when classes are imbalanced, which is why I also report PR-AUC, which concentrates on how well the malignant class is found (Saito and Rehmsmeier, 2015). Cross-validation (5 folds on the training set) gives the stable estimate; the test set gives one final unseen check.

Baseline	Accuracy	Recall	Precision	F1	ROC-AUC	PR-AUC
5-fold CV (mean)	0.9604	0.9235	0.9696	0.9454	0.9963	0.9951
Test set	0.9649	0.9286	0.9750	0.9512	0.9964	0.9935
Always 'benign' (test)	0.6316	0.0000	-	-	-	-

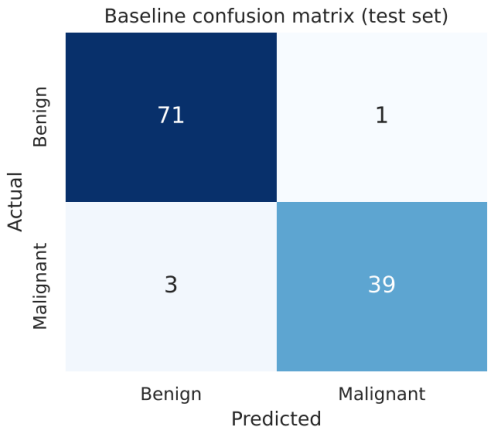


Figure 6: Baseline confusion matrix on the test set

The baseline ranks cases almost perfectly, but at the 0.5 threshold it **misses 3 of the 42 malignant cases**. For this brief that's the shortcoming to fix.

4.7 Retraining to fix the shortcoming

I made two changes, both chosen by cross-validation on the training data so that the test set stayed untouched:

- **Hyperparameters.** A grid search over C (0.01 to 100) and class_weight (None or balanced), scored on F2. The best was **C = 10 with balanced class weights** (CV F2 0.9683, up from 0.9320 for the baseline settings). Balanced weights make each malignant case count more in training.
- **Decision threshold.** From out-of-fold predicted probabilities I tested thresholds from 0.20 to 0.60. F2 peaked at **0.40** (out-of-fold recall 0.9765, precision 0.9486).

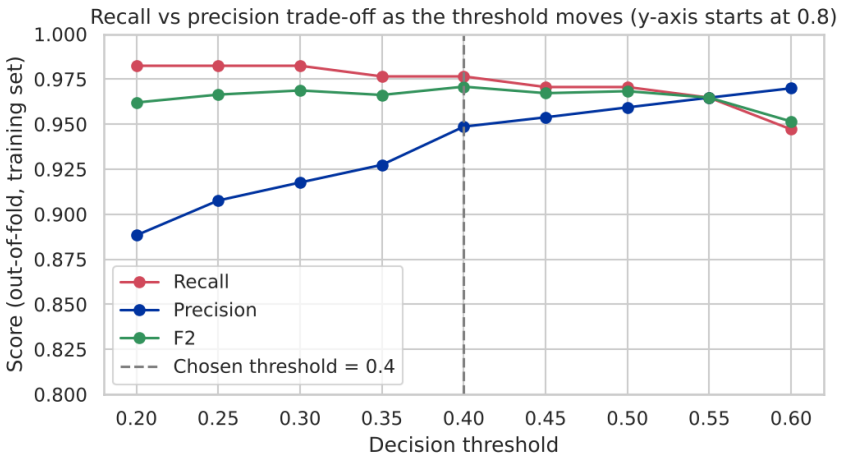


Figure 7: Recall, precision and F2 by threshold (note: y-axis starts at 0.8)

Test set	Recall	Precision	Specificity	F2	ROC-AUC	Missed cancers	False alarms
Baseline	0.9286	0.9750	0.9861	0.9375	0.9964	3	1
Tuned, threshold 0.5	0.9048	0.9500	0.9722	0.9135	0.9957	4	2
Final: tuned, threshold 0.4	0.9524	0.9524	0.9722	0.9524	0.9957	2	2

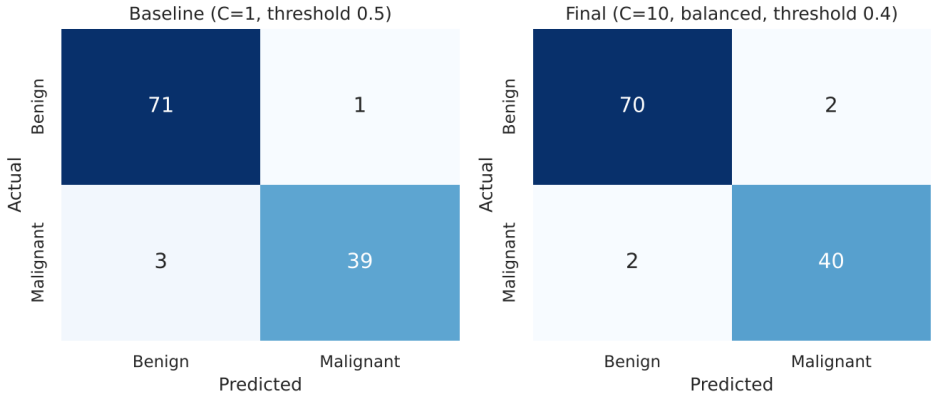


Figure 8: Confusion matrices before and after retraining

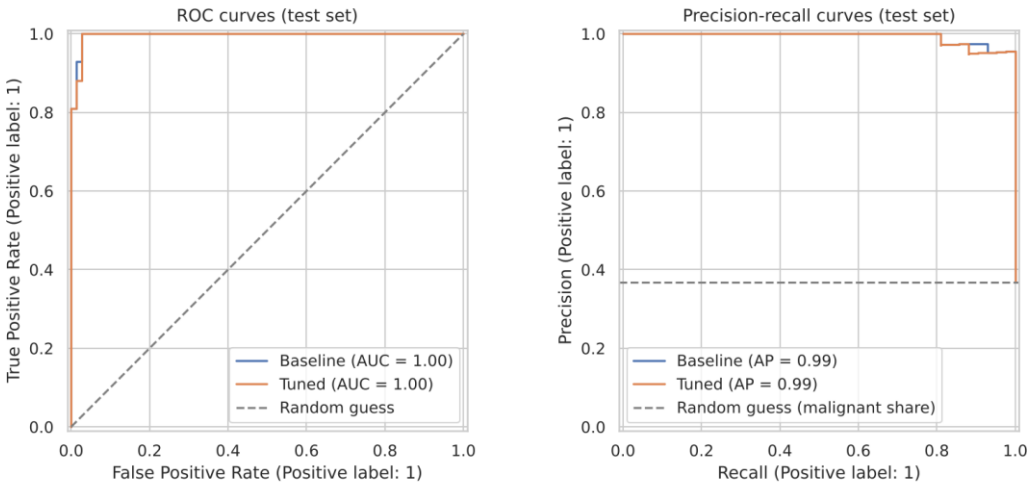


Figure 9: ROC and precision-recall curves on the test set

The final model catches **40 of 42** malignant cases instead of 39, at the cost of one extra false alarm. In the notebook's worked example, one malignant patient scores a probability of 0.4482: the baseline threshold would have missed them, the final model catches them.

Two honest caveats. First, at the default threshold the tuned model did slightly worse on this test set, even though it was better in cross-validation. With only 42 malignant test cases, each patient moves recall by 2.4 percentage points, so differences this small are within noise; that's why every choice was made on cross-validation. Second, ROC-AUC (0.9964 to 0.9957) and PR-AUC (0.9935 to 0.9924) barely moved, as Figure 9 shows. The ranking was already near perfect, so the gain comes from where the line is drawn, and that is a business decision about reviewer capacity.

4.8 What drives the model

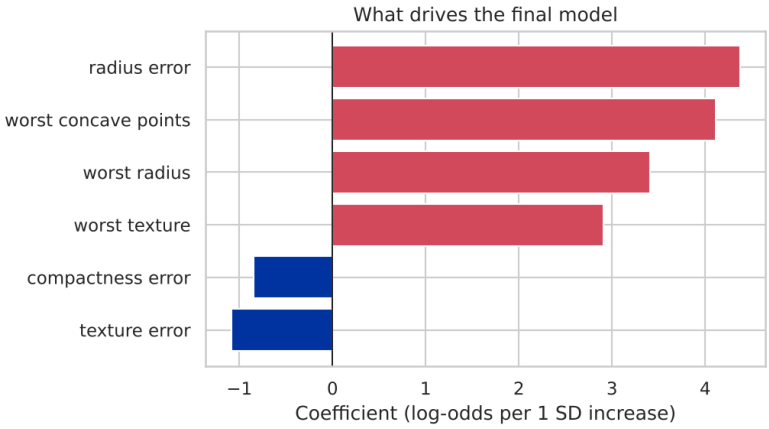


Figure 10: Final model coefficients (standardized features)

Converting the coefficients to odds ratios: one standard deviation more of radius error multiplies the odds of malignancy by about 79, worst concave points by 61, worst radius by 30 and worst texture by 18. Compactness error (0.43) and texture error (0.34) lower the odds once the other features are accounted for. In plain terms, **larger, more irregular and more variable nuclei point to cancer**, which agrees with what the EDA showed.

4.9 Conclusions and recommendations

- The final model catches 95.2% of malignant cases on unseen data, with 95.2% of its flags correct and a ROC-AUC of 0.9957.
- Use it to **fast-track** likely cancer claims to a clinical reviewer, never to decline a claim automatically. A human stays accountable for each decision.
- Set the threshold with the claims team: a lower threshold catches more cancers but sends more benign cases to reviewers.
- Retrain and validate on the scheme's own data, and extend to other cancers, before going live. Any real deployment must process member medical data in line with POPIA (South Africa, 2013).
- Firm up the estimates with repeated cross-validation or bootstrapped confidence intervals, since the test set holds only 42 malignant cases.

References

Hosmer, D.W., Lemeshow, S. and Sturdivant, R.X. (2013) *Applied logistic regression*. 3rd edn. Hoboken, NJ: John Wiley & Sons.

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An introduction to statistical learning: with applications in R*. 2nd edn. New York: Springer.

Muller, A.C. and Guido, S. (2016) *Introduction to machine learning with Python: a guide for data scientists*. Sebastopol, CA: O'Reilly Media.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, E. (2011) 'Scikit-learn: machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825-2830.

Saito, T. and Rehmsmeier, M. (2015) 'The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets', *PLoS ONE*, 10(3), e0118432. Available at: https://doi.org/10.1371/journal.pone.0118432 (Accessed: 21 October 2026).

Seabold, S. and Perktold, J. (2010) 'Statsmodels: econometric and statistical modeling with Python', *Proceedings of the 9th Python in Science Conference*. Austin, TX, 28 June - 3 July, pp. 92-96.

South Africa (2013) *Protection of Personal Information Act 4 of 2013*. Pretoria: Government Printer.

Street, W.N., Wolberg, W.H. and Mangasarian, O.L. (1993) 'Nuclear feature extraction for breast tumor diagnosis', *Biomedical Image Processing and Biomedical Visualization: Proceedings of SPIE 1905*. San Jose, CA, 1-4 February, pp. 861-870.

Wolberg, W., Mangasarian, O., Street, N. and Street, W. (1993) *Breast Cancer Wisconsin (Diagnostic)* [Dataset]. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5DW2B (Accessed: 21 October 2026).